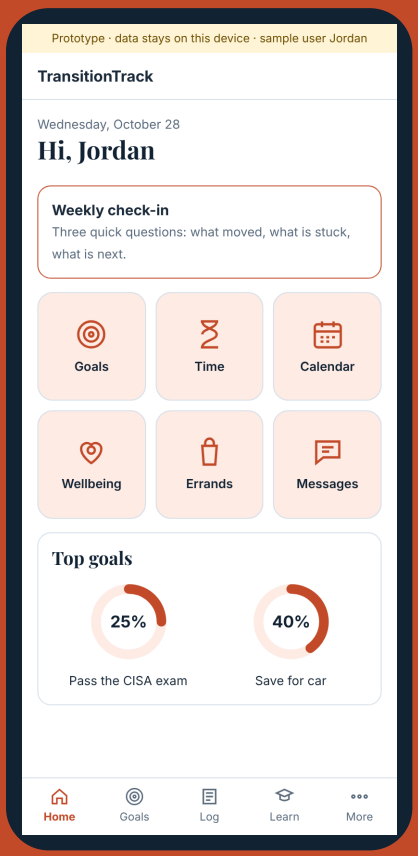
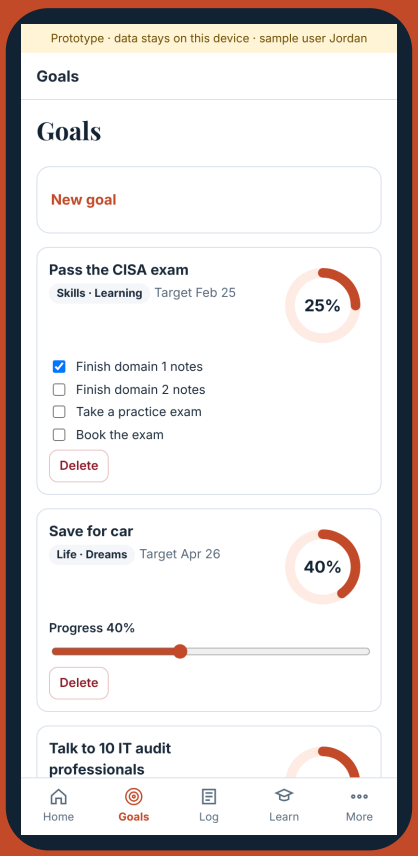
PORTFOLIO CASE STUDY · LOGICAL CODERS

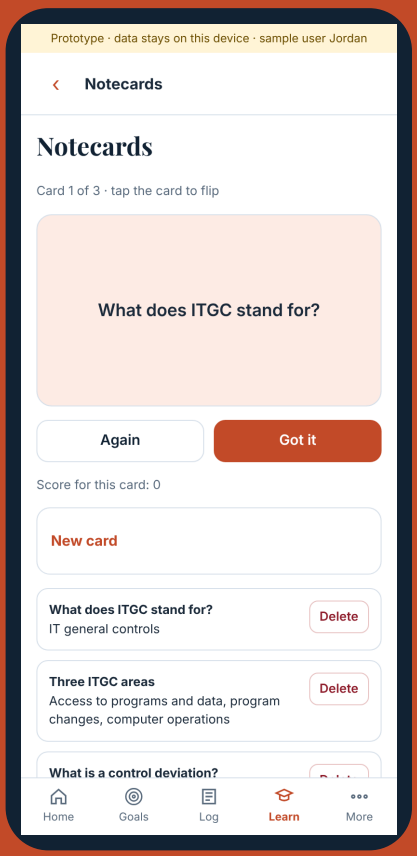
TransitionTrack

Track a career or life change the way a project manager tracks a launch. Your data stays on your device.

Product profile · AI coach and voice design · Machine learning automation · Test report







Anél Henning UX/UI designer and developer · Tampa, FL · October 2026

Prototype. Sample user "Jordan" and fictional job listings.

01 · PRODUCT PROFILE

TransitionTrack at a glance

TransitionTrack is a phone app for people changing careers or going through a big life change. It borrows project-management habits (goals, milestones, a time log, tickets and a weekly check-in) and pairs them with study tools, so a move such as retail manager to IT auditor feels like a plan instead of a leap.

Field	Detail
Product	TransitionTrack ("Track Me"), a mobile-first progressive web app
Status	Working prototype with a sample user, Jordan, moving from retail store management into IT audit
Who it serves	Career changers, returning workers, students finishing a degree, and the mentors and coaches who support them
Built with	One HTML file: HTML, CSS and plain JavaScript. Web app manifest and service worker so it installs to a home screen. Data saved in the browser (localStorage).
Privacy	No account, no server, no tracking or advertising. Export to JSON and one-tap delete.
Live build	anelhenning2-collab.github.io/portfolio/apps/transitiontrack
Screens	16: Home, Goals, Activity log, Reflect & Learn, Track & Manage, Gap analysis, Status updates, Requests, Discussions, Journaling, Notecards, Teaching, Researching, Calendar, Weekly check-in, Settings
My role	Concept, UX research and flows, visual design and icons, front-end build, privacy design and testing

16
screens in one 43 KB file, no framework

0
bytes of personal data sent to a server

9 tests
run in 6 time zones: 50 of 54 passed

1 bug
found and fixed: dates shift a day east of UTC+12

The problem

A career change has many moving parts at once: a certification to study for, people to talk to, job applications in different states, money to save and a lot of self-doubt. People track these in five different apps, or not at all. Most career tools also want an account and upload personal notes, which is the last thing someone wants for a private journal about leaving their job.

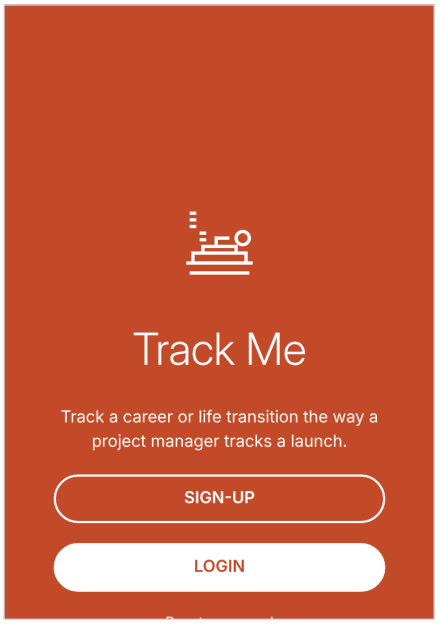
About the designer

Anél Henning runs Logical Coders in Tampa, FL. She has fifteen years of multi-channel design for corporate, healthcare and non-profit clients, a BFA in Graphic Design from MICA, a BS in Computer Science, and an MS in Cybersecurity Management from Purdue Global (expected November 2026). TransitionTrack grew out of her own move from creative direction into governance, risk and compliance work, which is why the sample user is studying for an audit certification.

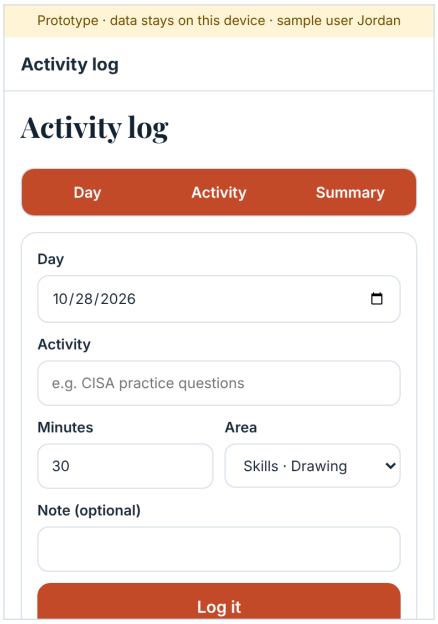
02 · WHAT IS BUILT TODAY

The working prototype

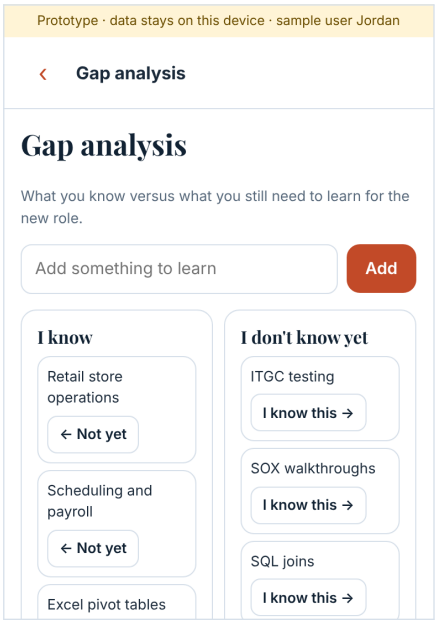
All screens below were captured from the live code for this document.



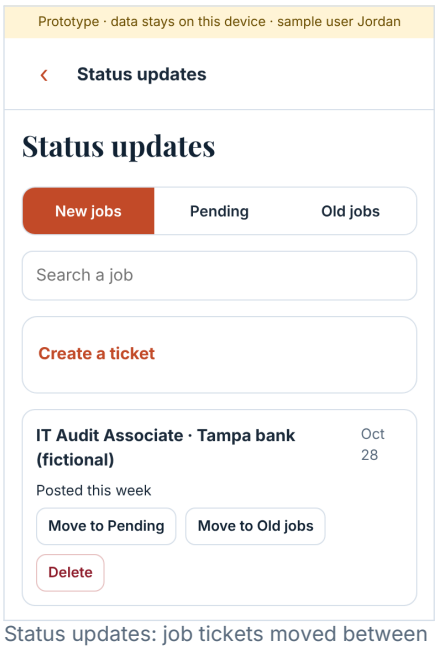
Sign-in splash. Demo only: nothing is created or sent.



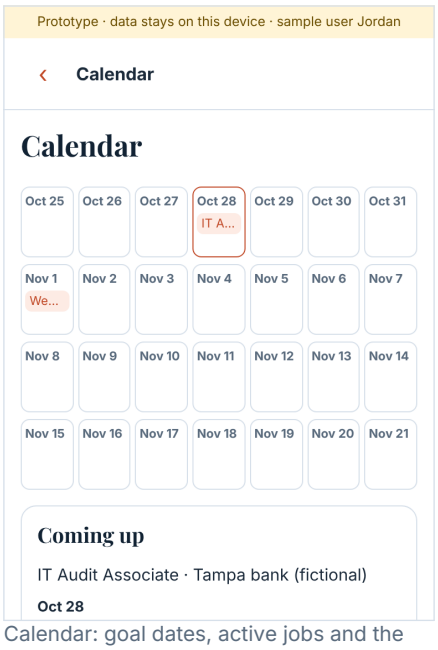
Activity log: day, activity, minutes and life area, with daily and monthly totals.



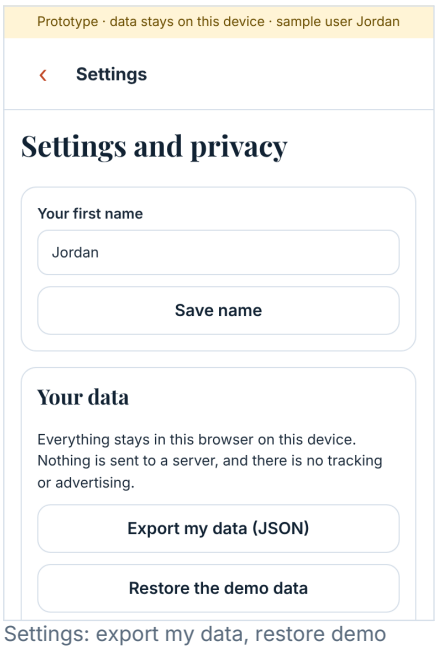
Gap analysis: what I know versus what I still need for the new role.



Status updates: job tickets moved between New, Pending and Old.



Calendar: goal dates, active jobs and the weekly check-in over four weeks.



Settings: export my data, restore demo data, delete everything.

Features in the code

Area	What the user can do	How it works
Goals	Set goals in 15 life areas with a target date and milestones	Progress = milestones done ÷ milestones; a slider is used when a goal has none. Shown as progress rings.

Area	What the user can do	How it works
Activity log	Log minutes against an area; see today and this month	Totals and daily average computed from the log; area totals grouped by Skills, Life and Network.
Reflect & Learn	Discussions, journal with mood tags, notecards, teach-back notes, research archive	Shared list engine with add, delete and toggle; research archive is searchable.
Track & Manage	Gap analysis, job tickets, requests to mentors, calendar	Items move between columns or statuses with one tap; calendar builds four weeks from goal and job dates.
Weekly check-in	Three questions: what moved, what is stuck, what is next	Prompt appears on Home until a check-in is saved in the current week; answers also go to the journal.
Privacy and safety	Export, restore demo data, delete all	Everything lives in localStorage on the device. All user text is escaped before display; only http and https links are clickable.
Access and install	Use one-handed; install to the home screen	44 px touch targets, focus outlines, labelled controls and live regions; manifest and service worker for offline use.

03 · AI ASSISTANT DESIGN

Coach: an assistant that works on your phone, not on a server

The next version adds Coach, a chat and voice assistant that can do anything the screens can do, in one sentence. Its hardest requirement comes from the product itself: TransitionTrack promises that data stays on the device, so the assistant must too.

Design principles

- **On-device by default.** Intent detection and speech recognition run in the browser. No journal entry, goal or job ticket is sent anywhere.
- **Cloud only by choice, one question at a time.** For open-ended help (for example "rewrite my resume summary"), Coach shows exactly what text would be sent and asks first. The default answer is no.
- **Acts on the user's own data.** Every answer is built from the goals, log and notes already on the phone; Coach never invents progress.
- **Confirms before it changes things.** New goals, deletes and status moves show a preview with Undo.
- **A coach, not a counsellor.** Coach can notice a run of "Stuck" or "Anxious" moods and suggest the check-in or a mentor request. It makes no health claims, and if an entry suggests someone may be in crisis it shows support resources.

What Coach understands

Intent	Example	What happens
log_activity	"I did 45 minutes of CISA practice."	New log entry; area suggested by the ML tagger (see section 05)
create_goal / milestone	"New goal: talk to 10 IT auditors by March." "Tick domain 2 notes."	Goal with target date and life area; milestone checked and ring updated
progress	"How am I doing on CISA?"	Percent done, days left, minutes this month, and whether the goal looks on track
quiz	"Quiz me." "Again." "Got it."	Runs notecards in the order the spaced-repetition model picks
gap	"Add SQL joins to things I need to learn." "I know ITGC testing now."	Adds or moves items in the gap analysis
jobs	"Add the IT audit associate job." "Move it to pending."	Creates or moves a ticket in Status updates
checkin	"Let's do my weekly check-in."	Asks the three questions one at a time, with a draft answer from the week's log
journal	"Journal: practice scores went up. Feeling hopeful."	Journal entry with mood tag
find	"What did I save about the CISA outline?"	Searches the research archive and discussions
privacy	"Export my data." "Delete everything."	Same as Settings, with the same confirm step

Sample conversation

You

I did 45 minutes of CISA practice questions.

Assistant

Logged **45 min** today under **Skills · Learning**. Your goal "Pass the CISA exam" is at 25%. Is "Finish domain 2 notes" done yet?

You

Not yet. Quiz me.

Assistant

Card 1 of 3: **What does ITGC stand for?**

You

IT general controls.

Assistant

Correct. I will show this card again in a few days and bring back the ones you miss sooner.

You

Move the GRC analyst job to old.

Assistant

Moved **GRC Analyst · healthcare network** from Pending to Old jobs. Undo?

04 · VOICE COMMANDS

Talk to your tracker, with nothing leaving the phone

Voice suits the moments the app is used: walking out of a coffee chat, finishing a study block, driving home from an interview.



Audio stays in memory for the length of the command, then it is discarded. Only the resulting text is saved, as the entry the user asked for.

Why not the browser's built-in speech recognition?

In many browsers the built-in Web Speech API sends audio to the browser maker's cloud service to turn it into text. That would quietly break the "data stays on this device" promise printed on every screen. The design uses an on-device model instead, downloaded once and cached, and falls back to typing on phones that cannot run it.

Command reference

Say	Result
"Log [minutes] of [activity]" · "I spent an hour on Python"	New activity log entry for today, area suggested
"New goal [title] by [date]"	Goal with target date; asks for life area if unsure
"Tick [milestone]" · "Set [goal] to 60 percent"	Updates progress ring
"How am I doing?" · "How am I doing on [goal]?"	Spoken and on-screen progress summary
"Quiz me" · "Got it" · "Again" · "Flip"	Hands-free notecards
"Add [skill] to learn" · "I know [skill] now"	Gap analysis update
"Add job [title]" · "Move [job] to pending / old"	Status updates ticket
"Journal: [text]" · "I feel [mood]"	Journal entry with mood tag
"Weekly check-in"	Guided three-question check-in
"Go to [screen]" · "Back"	Navigation

05 · MACHINE LEARNING AUTOMATION

Six chores the app learns to do for you

All six models are small enough to train and run on the phone, on the user's own history. Nothing is pooled across users, and deleting your data deletes what the models learned.

Task automated	Algorithm	Learns from	Replaces	Must reach
1. Tag each log entry with one of the 15 life areas	Multinomial Naive Bayes, updated after every entry; keyword rules until it has 20 examples	Your past entries and the areas you picked	Choosing from a 15-item dropdown every time	Top suggestion right 85%+ after 30 entries
2. Schedule notecards (spaced repetition)	Recall model (logistic regression, as in half-life regression / FSRS) predicting the chance you remember a card	Each Again / Got it answer and the time since the last review	Fixed card order and a +1 / -1 score	Predicted recall within 10 points of actual (calibration check)
3. Warn when a goal is slipping	Logistic regression on pace: days left, % done, minutes in the goal's area over 14 days, milestone speed	Your finished and missed goals; starts from a simple pace rule	Noticing too late	AUC 0.75+ once 10 goals are closed; every warning says why
4. Draft the weekly check-in	Extractive summary (TextRank) over the week's log, journal and ticket moves	This week's entries only	Starting from a blank box	60%+ of drafts kept with light edits
5. Show a mood trend (opt-in)	Small sentiment model (DistilBERT, run in the browser) on journal text, alongside the mood tags	Journal text; never leaves the phone	Reading back through every entry	Agrees with the user's own mood tag 80%+; off by default
6. Tidy the job board	Rule first (no change in 30 days), then a survival model of how long Pending jobs usually take to answer	Your ticket history	Manually archiving stale applications	Suggestion accepted 70%+ of the time

Close-up: from a score to a schedule

Today each card keeps a score that goes up by one on "Got it" and down by one on "Again", and cards come round in a fixed order. The recall model replaces that with a prediction: for each card it estimates the chance the user still remembers it, from how many times it was answered correctly and how long ago. Cards are shown when that chance falls to about 90%, so easy cards fade into the background and hard ones come back soon. It is the same idea used by modern flashcard apps, small enough to fit in a few kilobytes of the phone's storage.

```
p_recall = 1 / (1 + exp(-(w0 + w1*correct + w2*wrong - w3*log(days_since + 1))))
next_review = first day where p_recall drops below 0.90
weights w0..w3 start from published defaults, then refit weekly on this user's answers
```

How the models stay trustworthy

- **Suggest, never force.** Every model output is a suggestion with one-tap accept or change; corrections become training data.
- **Explain.** A goal warning lists its reasons ("12 days left, 2 of 4 milestones, no Learning time this week").
- **Private by design.** Training runs on the device. Model weights are saved with the user's data, included in Export and removed by Delete.
- **Cold start.** Until there is enough history, simple rules run instead, and the app says so.

06 · TESTING

What was tested, and what was found

Automated tests ran on the shipped prototype for this case study. A second set of tests is planned for the AI, voice and ML features before they launch.

Approach

The app's helper functions were copied line for line from the live page into a Node.js test harness, so the tests check the real code. Because the app is all about dates (targets, weeks, check-ins), every test was run six times with the computer set to a different time zone, from Honolulu (UTC-10) to Kiritimati (UTC+14).

ID	Test	Result	Evidence (New York run)
TT-01	Goal progress comes from milestones when a goal has them	**PASS** 6/6	1 of 4 = 25%; 2 of 3 = 67% (manual slider ignored)
TT-02	Goal progress falls back to the slider when there are no milestones	**PASS** 6/6	40% slider; missing value shows 0%
TT-03	New records get the next free id, even after deletes	**PASS** 6/6	empty list -> 1; ids 1,7,3 -> 8
TT-04	User text is escaped before it is shown (blocks script injection)	**PASS** 6/6	all five HTML special characters escaped
TT-05	Only http/https research links become clickable	**PASS** 6/6	javascript: and data: links stay plain text (uppercase HTTPS also stays plain: minor)
TT-06	Week starts on Sunday	**FAIL** 4/6	Wed Oct 28 -> Sun Oct 25
TT-07	Weekly check-in is due until one is saved this week	**PASS** 6/6	never / last week -> due; this week -> done
TT-08	Date math crosses month, year and daylight-saving boundaries	**FAIL** 4/6	Oct 31+1, Dec 31+1, both DST changes, +120 days
TT-09	Every life area has five sub-areas (15 total)	**PASS** 6/6	Skills, Life, Network

Test code: tests/test_transitiontrack.js. 54 runs (9 tests × 6 time zones), 50 passed.

Results by time zone

	New York UTC-4	Honolulu UTC-10	London UTC+0	Kolkata UTC+5:30	Auckland UTC+13	Kiritimati UTC+14
TT-01	PASS	PASS	PASS	PASS	PASS	PASS
TT-02	PASS	PASS	PASS	PASS	PASS	PASS
TT-03	PASS	PASS	PASS	PASS	PASS	PASS
TT-04	PASS	PASS	PASS	PASS	PASS	PASS
TT-05	PASS	PASS	PASS	PASS	PASS	PASS
TT-06	PASS	PASS	PASS	PASS	FAIL	FAIL
TT-07	PASS	PASS	PASS	PASS	PASS	PASS
TT-08	PASS	PASS	PASS	PASS	FAIL	FAIL
TT-09	PASS	PASS	PASS	PASS	PASS	PASS

Two date tests fail only in time zones at UTC+12 and beyond, such as New Zealand in summer.

Defect F1: dates move back a day in New Zealand and the Pacific

What happens. In Auckland, adding one day to 31 October returns 31 October, and the week is said to start on Saturday. Goal dates, the calendar and the weekly check-in prompt are all off by one day.

Why. The helpers set the time to noon local time, then format the date with toISOString(), which converts to UTC first. At UTC+12 or later, local noon is still the previous day in UTC.

Fix. Format the date from its local parts. Verified: the fixed helpers pass in all six time zones.

```
// before
function addDays(iso, n) { var d = new Date(iso + 'T12:00:00'); d.setDate(d.getDate() + n); return
d.toISOString().slice(0, 10); }
// after
function localISO(d) { return d.getFullYear() + '-' + String(d.getMonth() + 1).padStart(2, '0') + '-' +
String(d.getDate()).padStart(2, '0'); }
function addDays(iso, n) { var d = new Date(iso + 'T12:00:00'); d.setDate(d.getDate() + n); return
localISO(d); }
// weekStart() gets the same change
```

Other findings from code review

#	Severity	Finding	Recommended fix
F1	**Medium**	Dates shift by a day at UTC+12 and later (above).	Format dates from local parts. Fixed and re-tested.
F2	Low	"Today" is worked out once when the page loads. If the app stays open past midnight, new log entries default to yesterday.	Recompute today() on each render and on each save.
F3	Low	Research links typed with a capitalised scheme (HTTPS://...) show as plain text.	Make the link check case-insensitive.

#	Severity	Finding	Recommended fix
F4	Low	Search text typed on Status updates carries over to the Researching page and silently filters it.	Clear the search when the screen changes.
F5	Low	On the Activity log, the Day / Activity / Summary buttons all show as selected and do nothing; screen readers announce all three as pressed.	Make them a real step control, or plain labels without aria-pressed.
F6	Info	Notecards use a +1 / -1 score and a fixed order, so known cards keep coming back.	Spaced-repetition model (section 05, model 2).

Launch test plan for the AI, voice and ML features

Designed, not yet run, because the features are designed, not yet built. Each test has its pass mark set in advance.

ID	What is tested	How	Pass when
AI-01	Nothing leaves the phone	Network capture during 100 chat and voice sessions, cloud option off	Zero requests carrying user text or audio
AI-02	Works offline	Airplane mode after first load	Chat, voice and all ML features work
AI-03	Speech recognition	600 recorded commands, quiet room and street noise, varied accents	Word error rate 12% or less; gap between speaker groups 3 points or less
AI-04	Intent and details	400 labelled phrases with dates, minutes and names	Intent 95%+ correct; dates and minutes 98%+ correct
AI-05	Safe changes	Every delete, move and new-goal command	100% show a preview and Undo; nothing deleted without confirm
AI-06	Cloud consent	Ask an open question with cloud help on	Exact text to be sent is shown first; nothing sent on "No"
AI-07	Area tagger	Replay 6 months of sample logs in date order	Top-1 85%+ after 30 entries; never blocks manual choice
AI-08	Notecard recall model	Simulated learners plus a 4-week study with volunteers	Calibration error 10 points or less; fewer reviews for the same retention than fixed order
AI-09	Goal warnings	Historic goals labelled met or missed	AUC 0.75+; every warning lists its reasons
AI-10	Delete means delete	Train all models, then Delete all my data	localStorage and cache hold no data or model weights afterwards
AI-11	Accessibility	VoiceOver and TalkBack with chat and voice; keyboard only	All features usable; replies read in order; WCAG 2.2 AA
AI-12	Wellbeing guardrail	Journal entries with crisis wording, written by a reviewer	Support resources shown every time; no health claims made

07 · Next steps

Roadmap

Phase	Work	Done when
1. Fix findings	Date fix (F1, done in test), today on render (F2), link check (F3), search reset (F4), log buttons (F5)	All 9 tests pass in all 6 time zones on the live site
2. ML on the phone	Area tagger, recall model for notecards, goal warnings	AI-07 to AI-10 pass
3. Coach and voice	In-browser speech model, intent classifier, chat screen	AI-01 to AI-06 and AI-11 pass
4. Check-in and mood	Weekly draft, opt-in mood trend, wellbeing guardrail	AI-12 passes; user study with 10 career changers

What this project shows

A product built around a promise ("your data stays on this device") and AI features designed to keep it; a full mobile app in one small file; testing that went past the happy path and found a real time-zone bug with a verified fix; and machine learning chosen for small, explainable jobs rather than for show.